EXERCISE 10 SEQUENCES AND SYNONYMS

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PART A — SEQUENCES

Create sequence starting from 1 increment by 1

```
SQL> CREATE SEQUENCE seq_student_24BAI1150
  2  START WITH 1
  3  INCREMENT BY 1;

Sequence created.
```

Create sequence for course IDs starting from 500

```
SQL> CREATE SEQUENCE seq_course_24BAI1150
  2  START WITH 500
  3  INCREMENT BY 1;

Sequence created.
```

Insert record into STUDENT using sequence

```
SQL> INSERT INTO STUDENT_24BAI1150
  2  VALUES(seq_student_24BAI1150.NEXTVAL,'Neha','CSE','Chennai');

1 row created.
```

Display next value of sequence

```
SQL> SELECT seq_student_24BAI1150.NEXTVAL FROM dual;

   NEXTVAL
----------
         2
```

Create sequence with increment 10

```
SQL> CREATE SEQUENCE seq_inc10_24BAI1150
  2  START WITH 10
  3  INCREMENT BY 10;

Sequence created.
```

Create sequence with MAXVALUE 10000

```
SQL> CREATE SEQUENCE seq_max_24BAI1150
  2  START WITH 1
  3  INCREMENT BY 1
  4  MAXVALUE 10000
  5  NOCYCLE;

Sequence created.
```

Create cycling sequence

```
SQL> CREATE SEQUENCE seq_cycle_24BAI1150
  2  START WITH 1
  3  INCREMENT BY 1
  4  MAXVALUE 10
  5  CYCLE
  6  NOCACHE;

Sequence created.
```

Use sequence in bulk insert

```
SQL> INSERT INTO STUDENT_24BAI1150
  2  VALUES(seq_student_24BAI1150.NEXTVAL,'Rahul','IT','Chennai');

1 row created.

SQL>
SQL> INSERT INTO STUDENT_24BAI1150
  2  VALUES(seq_student_24BAI1150.NEXTVAL,'Priya','ECE','Madurai');

1 row created.
```

Find current value of sequence

```
SQL> SELECT seq_student_24BAI1150.CURRVAL FROM dual;

   CURRVAL
----------
         4
```

Drop existing sequence

```
SQL> DROP SEQUENCE seq_student_24BAI1150;

Sequence dropped.
```

All the sequence in my schema

```
SQL> SELECT sequence_name
  2  FROM user_sequences;

SEQUENCE_NAME
-----------------------------
SEQ_COURSE_24BAI1150
SEQ_CYCLE_24BAI1150
SEQ_INC10_24BAI1150
SEQ_MAX_24BAI1150
```

PART B — SYNONYMS

Create synonym for STUDENT table

```
SQL> CREATE SYNONYM stud_24BAI1150
  2  FOR STUDENT_24BAI1150;

Synonym created.
```

Create synonym for COURSE table

```
SQL> CREATE SYNONYM course_syno_24BAI1150
  2  FOR COURSE_24BAI1150;

Synonym created.
```

Use synonym in SELECT query

```
SQL> SELECT * FROM stud_24BAI1150;

STUDENT_ID STUDENT_NAME
========== ==================================================
DEPARTMENT
==================================================
CITY
==================================================
       101 Arjun
CSE
Chennai

       102 Meera
ECE
Madurai

STUDENT_ID STUDENT_NAME
========== ==================================================
DEPARTMENT
==================================================
CITY
==================================================

       103 Rohan
IT
Coimbatore

       104 Kavya
CSE

STUDENT_ID STUDENT_NAME
========== ==================================================
DEPARTMENT
==================================================
CITY
==================================================
Trichy

       105 Divya
ECE
Salem

         1 Neha

STUDENT_ID STUDENT_NAME
========== ==================================================
DEPARTMENT
==================================================
CITY
==================================================
CSE
Chennai

         3 Rahul
IT
Chennai


STUDENT_ID STUDENT_NAME
========== ==================================================
DEPARTMENT
==================================================
CITY
==================================================
         4 Priya
ECE
Madurai


8 rows selected.
```

Create synonym for sequence

```
SQL> CREATE SYNONYM stud_seq_24BAI1150
  2  FOR seq_student_24BAI1150;

Synonym created.
```

Create synonym for a view

```
SQL> CREATE VIEW student_city_view_24BAI1150 AS
  2  SELECT student_name, city
  3  FROM STUDENT_24BAI1150;

View created.
```

NEXT PAGE

Replace table name with synonym in query

```
SQL> SELECT * FROM stud_24BAI1150;

STUDENT_ID STUDENT_NAME
---------- ------------------------------------------------
DEPARTMENT
------------------------------------------------
CITY
------------------------------------------------
       101 Arjun
CSE
Chennai

       102 Meera
ECE
Madurai

STUDENT_ID STUDENT_NAME
---------- ------------------------------------------------
DEPARTMENT
------------------------------------------------
CITY
------------------------------------------------

       103 Rohan
IT
Coimbatore

       104 Kavya
CSE

STUDENT_ID STUDENT_NAME
---------- ------------------------------------------------
DEPARTMENT
------------------------------------------------
CITY
------------------------------------------------
Trichy

       105 Divya
ECE
Salem

         1 Neha

STUDENT_ID STUDENT_NAME
---------- ------------------------------------------------
DEPARTMENT
------------------------------------------------
CITY
------------------------------------------------
CSE
Chennai

         3 Rahul
IT
Chennai


STUDENT_ID STUDENT_NAME
---------- ------------------------------------------------
DEPARTMENT
------------------------------------------------
CITY
------------------------------------------------
         4 Priya
ECE
Madurai


8 rows selected.
```

Drop a synonym

```
SQL> DROP SYNONYM stud_24BAI1150;

Synonym dropped.
```

Create PUBLIC synonym (Theory)

```
SQL> CREATE PUBLIC SYNONYM stud_pub_24BAI1150
  2  FOR STUDENT_24BAI1150;
CREATE PUBLIC SYNONYM stud_pub_24BAI1150
*
ERROR at line 1:
ORA-01031: insufficient privileges
```

Advantage of synonyms

1. Simplifies long table names
2. Hides schema names
3. Provides location transparency
4. Improves security abstraction
5. Makes queries easier to read

List all synonyms in database

```
SQL> SELECT synonym_name
  2  FROM user_synonyms;

SYNONYM_NAME
------------------------------
STUD_SEQ_24BAI1150
COURSE_SYNO_24BAI1150
```